UCL COMP0078: Supervised Learning
Problem Sheet
Support Vector Machines

Solutions by Stephen Pasteris & Antonin Schrab

If you find any mistakes/typos, please contact sl-support@cs.ucl.ac.uk.

Overview of Problems

- Problem 1.1: Diagram visualisation of hard margin SVM.
- Problem 1.2: Diagram visualisation of the effect of removing points for SVM (support vectors and others).
- Problem 1.3: Lecture slide 17 explains that minimising the regularised misclassification loss is NP-hard, that this optimisation can be simplified by using the hinge loss instead. This problem shows that solving an optimisation problem (P3 in lecture slides 18) is equivalent to minimising the regularised hinge loss.
- Problems 2.1, 2.2 & 2.3: Classical example of using the method of Lagrange multipliers to solve the SVM optimisation problem.

Problem 1.1

Describe the criterion used by hard margin Support Vector Machines to choose a separating hyperplane for a linearly separable dataset, illustrating with a diagram indicating the margin and the support vectors.

Solution 1.1

The criterion is to choose the separating hyperplane (i.e. an affine plane which has all positive examples on one side and all negative examples on the other) that maximises the distance of the nearest example to it.

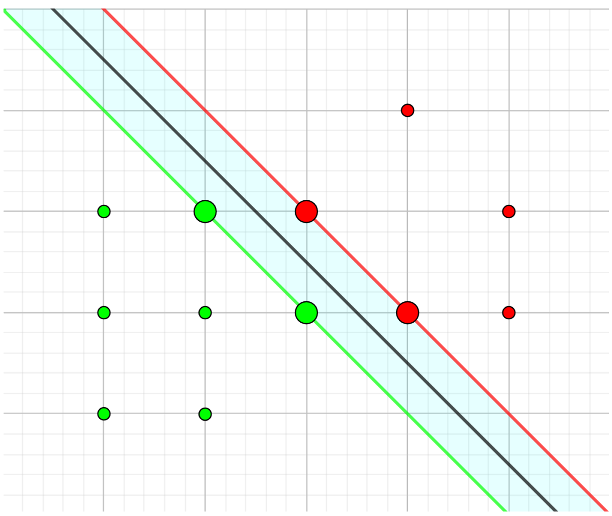

Figure 1: Support vectors are displayed in bigger size.

Problem 1.2

For a hard-margin SVM. If we remove one of

1. *the examples which is a support vector from the training set,*
2. *the examples which is not a support vector from the training set,*

and retrain with out that example. How does the maximum margin change?

Solution 1.2

If we remove one example which is a support vector (1.2.1), then either the maximum margin remains the same (see Figure 2) or it increases (see Figure 3), the margin never decreases.
If we remove examples which are not support vectors (1.2.2), then the optimal hyperplane with its maximum margin remains the same (see Figure 4).

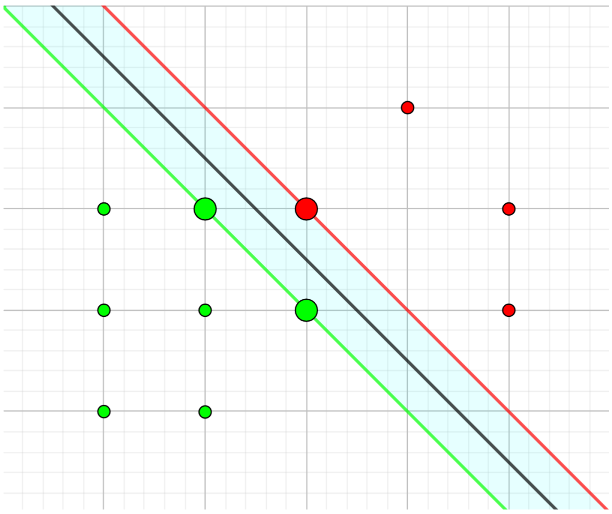

Figure 2: Removing one support vector does not necessarily change the maximum margin.

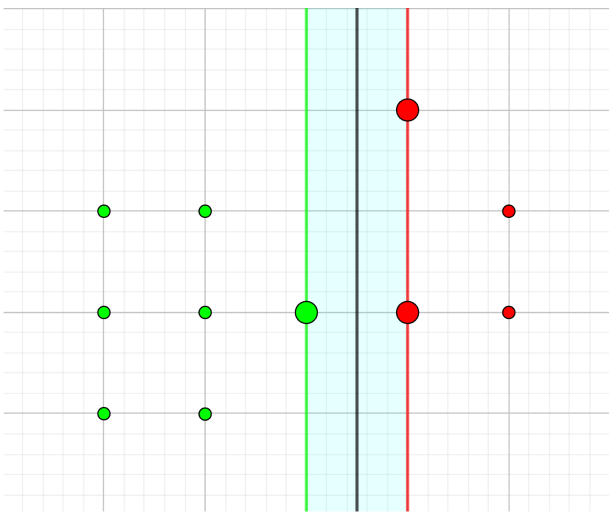

Figure 3: Removing one support vector can increase the maximum margin.

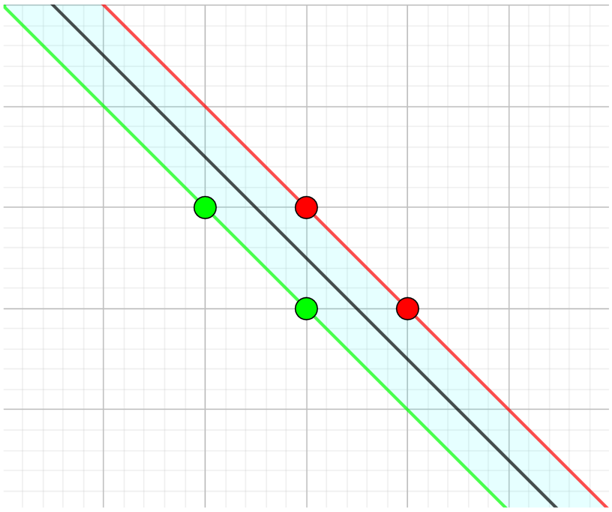

Figure 4: Removing non support vector does not change the maximum margin.

Problem 1.3

Consider the optimisation given by

$$\begin{aligned}
&\min_{\boldsymbol{w},\xi} \quad \|\boldsymbol{w}\| + C\sum_{i=1}^{m}\xi_i \\
&\text{subject to} \quad y_i\langle \boldsymbol{w}, \boldsymbol{x}_i\rangle \geq 1-\xi_i,\ \xi_i \geq 0,\ i=1,\ldots,m
\end{aligned}$$

Which part of the optimisation corresponds to the regulariser? What is the loss function incorporated into the optimisation?

Solution 1.3

We can rewrite the optimisation problem as

$$\begin{aligned}
&\min_{\boldsymbol{w},\xi} \quad \sum_{i=1}^{m}\xi_i + C^{-1}\|\boldsymbol{w}\| \\
&\text{subject to} \quad \xi_i \geq 1 - y_i\langle \boldsymbol{w}, \boldsymbol{x}_i\rangle,\ \xi_i \geq 0,\ i=1,\ldots,m
\end{aligned}$$

which we rewrite as

$$\begin{aligned}
&\min_{\boldsymbol{w},\xi} \quad \sum_{i=1}^{m}\xi_i + C^{-1}\|\boldsymbol{w}\| \\
&\text{subject to} \quad \xi_i \geq \max\Big(0, 1 - y_i\langle \boldsymbol{w}, \boldsymbol{x}_i\rangle\Big),\ i=1,\ldots,m
\end{aligned}$$

or equivalently

$$\min_{\boldsymbol{w}} \quad \sum_{i=1}^{m}\max\Big(0, 1 - y_i\langle \boldsymbol{w}, \boldsymbol{x}_i\rangle\Big) + C^{-1}\|\boldsymbol{w}\|$$

that is

$$\min_{\boldsymbol{w}} \quad \sum_{i=1}^{m}V_{\text{hinge}}\Big(y_i, \langle \boldsymbol{w}, \boldsymbol{x}_i\rangle\Big) + C^{-1}\|\boldsymbol{w}\|.$$

So, the loss function incorporated into the optimisation is the hinge loss, the term $\|\boldsymbol{w}\|$ corresponds to the regulariser with regularisation coefficient C^{-1}.

Problem 2.1

Assume that the set $S = \{(\boldsymbol{x}_i, y_i)\}_{i=1}^m \subset \mathbb{R}^2 \times \{-1, 1\}$ *of binary examples is strictly linearly separable by a line going through the origin, that is, there exists a vector* $\boldsymbol{w} \in \mathbb{R}^2$ *such that the linear function* $f(\boldsymbol{x}) = \boldsymbol{w}^\top \boldsymbol{x}$, $\boldsymbol{x} \in \mathbb{R}^2$ *has the property that* $y_i f(\boldsymbol{x}_i) > 0$ *for every* $i = 1, \cdots, m$. *Consider the optimisation problem (linearly separable SVM):*

$$\begin{aligned} \text{P1}: \quad & \min_{\boldsymbol{w}} \quad \frac{1}{2}\boldsymbol{w}^\top \boldsymbol{w} \\ & \text{subject to} \quad y_i \boldsymbol{w}^\top \boldsymbol{x}_i \geq 1, \; i = 1, \cdots, m \end{aligned}$$

Argue that the above problem has a unique solution. Describe the geometric meaning of this solution.

Solution 2.1

Solution exists. Since the data is strictly linearly separable by a line going through the origin (by assumption), there exists some $\bar{\boldsymbol{w}} \in \mathbb{R}^2$ satisfying $y_i \bar{\boldsymbol{w}}^\top \boldsymbol{x}_i \geq 1$ for $i = 1, \ldots, m$ (which is the constraint of P1). Hence, the optimisation problem reduces to

$$\begin{aligned} & \min_{\boldsymbol{w}} \quad \frac{1}{2}\boldsymbol{w}^\top \boldsymbol{w} \\ & \text{subject to} \quad \boldsymbol{w}^\top \boldsymbol{w} \leq \bar{\boldsymbol{w}}^\top \bar{\boldsymbol{w}} \text{ and } y_i \boldsymbol{w}^\top \boldsymbol{x}_i \geq 1, \; i = 1, \cdots, m \end{aligned}$$

which is optimised over a compact set (it was not the case for the original P1 formulation). Hence, it is guaranteed that a solution exists.

Solution is unique. The function $\frac{1}{2}\boldsymbol{w}^\top \boldsymbol{w}$ is strictly convex since $\frac{\partial^2}{\partial \boldsymbol{w}^2}\left(\frac{1}{2}\boldsymbol{w}^\top \boldsymbol{w}\right) = \frac{\partial}{\partial \boldsymbol{w}}(\boldsymbol{w}) = 1 > 0$. The ball $\{\boldsymbol{w} : \|\boldsymbol{w}\| \leq \|\bar{\boldsymbol{w}}\|\}$ of radius $\|\bar{\boldsymbol{w}}\|$ is a convex set. Also, for all $i = 1, \ldots, m$ we have that the set $\{\boldsymbol{w} : y_i \boldsymbol{w}^\top \boldsymbol{x}_i \geq 1\}$ is a half space which is a convex set. Since the intersection of a collection of convex sets is itself convex we then have that the set $\{\boldsymbol{w} : \|\boldsymbol{w}\| \leq \|\bar{\boldsymbol{w}}\|, \, y_i \boldsymbol{w}^\top \boldsymbol{x}_i \geq 1, i = 1, \ldots, m\}$ is convex, and from the problem description is non-empty. Our problem is then to minimise a strictly convex function over a non-empty convex compact set which always has an unique solution.

Geometric meaning. The geometric meaning of the solution is that it is the separating hyperplane, going through the origin, of which the distance of the nearest example to it is maximised.

Remark. Note that, using the usual bias trick of extending $\widetilde{\boldsymbol{w}} = (\boldsymbol{w}, b)$ and $\widetilde{\boldsymbol{x}} = (\boldsymbol{x}, 1)$, leads to

$$\begin{aligned} & \min_{\boldsymbol{w}, b} \quad \frac{1}{2}\left(\boldsymbol{w}^\top \boldsymbol{w} + b^2\right) \\ & \text{subject to} \quad y_i(\boldsymbol{w}^\top \boldsymbol{x}_i + b) \geq 1, \; i = 1, \cdots, m, \end{aligned}$$

and not to the usual SVM problem

$$\begin{aligned} & \min_{\boldsymbol{w}, b} \quad \frac{1}{2}\boldsymbol{w}^\top \boldsymbol{w} \\ & \text{subject to} \quad y_i(\boldsymbol{w}^\top \boldsymbol{x}_i + b) \geq 1, \; i = 1, \cdots, m. \end{aligned}$$

This is not desired as we want to maximize the margin (equivalently minimise $\|\boldsymbol{w}\|_2$) and the bias term b does not affect the margin. This bias simply shifts the hyperplane vertically, and there is no reason to encourage the hyperplane to be close to the origin (i.e. to minimise b).

Problem 2.2

Show that the vector $\boldsymbol{w}$ *solving problem* P1 *has the form* $\boldsymbol{w} = \sum_{i=1}^m c_i y_i \boldsymbol{x}_i$ *where* $c_1, \ldots, c_m$ *are some nonnegative coefficients. [HINT: use the method of Lagrange multipliers]*

Solution 2.2

Using the method of Lagrange multipliers, the optimisation problem P1 is equivalent to

$$\text{P1}^*: \quad \max_{\boldsymbol{c}\geq \boldsymbol{0}} \ \min_{\boldsymbol{w}} \ L(\boldsymbol{w}, \boldsymbol{c})$$

where the Lagrangian is

$$L(\boldsymbol{w}, \boldsymbol{c}) \ = \ \frac{1}{2}\boldsymbol{w}^\top \boldsymbol{w} - \sum_{i=1}^{m} c_i \big(y_i \boldsymbol{w}^\top \boldsymbol{x}_i - 1\big)$$

We first solve $\min_{\boldsymbol{w}} \ L(\boldsymbol{w}, \boldsymbol{c})$ by solving

$$\boldsymbol{0} = \nabla_{\boldsymbol{w}} L = \boldsymbol{w} - \sum_{i=1}^{m} c_i y_i \boldsymbol{x}_i$$

which gives us

$$\boldsymbol{w} = \sum_{i=1}^{m} c_i y_i \boldsymbol{x}_i.$$

The optimisation problem P1* becomes

$$\text{P1}^{**}: \quad \max_{\boldsymbol{c}\geq \boldsymbol{0}} \ L\left(\sum_{i=1}^{m} c_i y_i \boldsymbol{x}_i, \boldsymbol{c}\right)$$

Problem 2.3

Show that the coefficients $c_1, \ldots, c_m$ in the above formula solve the optimization problem

$$\text{P2}: \qquad \max \left\{ -\frac{1}{2} \sum_{i=1}^{m}\sum_{j=1}^{m} c_i c_j y_i y_j \boldsymbol{x}_i{}^\top \boldsymbol{x}_j + \sum_{i=1}^{m} c_i : c_j \geq 0, \ j = 1, \cdots, m \right\},$$

that is

$$\text{P2}: \quad \max_{\boldsymbol{c}\geq \boldsymbol{0}} \ -\frac{1}{2} \sum_{i=1}^{m}\sum_{j=1}^{m} c_i c_j y_i y_j \boldsymbol{x}_i{}^\top \boldsymbol{x}_j + \sum_{i=1}^{m} c_i.$$

Finally, if $(c_1, \ldots, c_m)$ is the solution to this problem and $\boldsymbol{w}$ is the solution to problem P1, argue that $\boldsymbol{w}^\top \boldsymbol{w} = \sum_{i=1}^{m} c_i$.

Solution 2.3

From Solution 2.2, we have the optimisation problem

$$\text{P1}^{**}: \quad \max_{\boldsymbol{c}\geq\boldsymbol{0}} \; L\left(\sum_{i=1}^{m} c_i y_i \boldsymbol{x}_i, \boldsymbol{c}\right)$$

where, for $\boldsymbol{w} = \sum_{i=1}^{m} c_i y_i \boldsymbol{x}_i$, we have

$$\begin{aligned}
L(\boldsymbol{w},\boldsymbol{c}) &= \frac{1}{2}\boldsymbol{w}^\top\boldsymbol{w} - \sum_{i=1}^{m} c_i\big(y_i\boldsymbol{w}^\top\boldsymbol{x}_i - 1\big) \\
&= \frac{1}{2}\boldsymbol{w}^\top\boldsymbol{w} - \sum_{i=1}^{m} c_i y_i\boldsymbol{w}^\top\boldsymbol{x}_i + \sum_{i=1}^{m} c_i \\
&= \frac{1}{2}\left(\sum_{i=1}^{m} c_i y_i\boldsymbol{x}_i\right)^\top\left(\sum_{j=1}^{m} c_j y_j\boldsymbol{x}_j\right) - \sum_{i=1}^{m} c_i y_i\left(\sum_{j=1}^{m} c_j y_j\boldsymbol{x}_j\right)^\top \boldsymbol{x}_i + \sum_{i=1}^{m} c_i \\
&= \frac{1}{2}\left(\sum_{i=1}^{m} c_i y_i\boldsymbol{x}_i^\top\right)\left(\sum_{j=1}^{m} c_j y_j\boldsymbol{x}_j\right) - \sum_{i=1}^{m} c_i y_i\left(\sum_{j=1}^{m} c_j y_j\boldsymbol{x}_j^\top\right) \boldsymbol{x}_i + \sum_{i=1}^{m} c_i \\
&= \frac{1}{2}\sum_{j=1}^{m}\sum_{j=1}^{m} c_i y_i\boldsymbol{x}_i^\top c_j y_j\boldsymbol{x}_j - \sum_{i=1}^{m}\sum_{j=1}^{m} c_i y_i c_j y_j \boldsymbol{x}_j^\top\boldsymbol{x}_i + \sum_{i=1}^{m} c_i \\
&= \frac{1}{2}\sum_{i=1}^{m}\sum_{j=1}^{m} c_i c_j y_i y_j\boldsymbol{x}_i^\top\boldsymbol{x}_j - \sum_{i=1}^{m}\sum_{j=1}^{m} c_i c_j y_i y_j \boldsymbol{x}_i^\top\boldsymbol{x}_j + \sum_{i=1}^{m} c_i \\
&= -\frac{1}{2}\sum_{i=1}^{m}\sum_{j=1}^{m} c_i c_j y_i y_j\boldsymbol{x}_i^\top\boldsymbol{x}_j + \sum_{i=1}^{m} c_i.
\end{aligned}$$

The problem P1^{**} becomes

$$\text{P2}: \quad \max_{\boldsymbol{c}\geq\boldsymbol{0}} \; -\frac{1}{2}\sum_{i=1}^{m}\sum_{j=1}^{m} c_i c_j y_i y_j\boldsymbol{x}_i^\top\boldsymbol{x}_j + \sum_{i=1}^{m} c_i,$$

which proves the first part of Problem 2.3.

Now, to solve P2 we find $\boldsymbol{c}\geq\boldsymbol{0}$ which maximizes

$$-\frac{1}{2}\sum_{i=1}^{m}\sum_{j=1}^{m} c_i c_j y_i y_j\boldsymbol{x}_i^\top\boldsymbol{x}_j + \sum_{i=1}^{m} c_i \;=\; -\frac{1}{2}\boldsymbol{c}^\top\boldsymbol{A}\boldsymbol{c} + \boldsymbol{c}^\top\boldsymbol{1}$$

where $\boldsymbol{A}_{i,j} = y_i y_j \boldsymbol{x}_i^\top\boldsymbol{x}_j$ and $\boldsymbol{1} = (1,\ldots,1)^\top$. To solve this maximisation we want to take the gradient and set it equal to 0 giving

$$\begin{aligned}
\boldsymbol{0} &= \nabla_{\boldsymbol{c}}\Big(-\frac{1}{2}\boldsymbol{c}^\top\boldsymbol{A}\boldsymbol{c} + \boldsymbol{c}^\top\boldsymbol{1}\Big) \\
\boldsymbol{0} &= -\boldsymbol{A}\boldsymbol{c} + \boldsymbol{1} \\
\boldsymbol{A}\boldsymbol{c} &= \boldsymbol{1}
\end{aligned}$$

that is, for $i = 1,\ldots,m$,

$$\sum_{j=1} c_j y_i y_j \boldsymbol{x}_i^\top\boldsymbol{x}_j = \big(\boldsymbol{A}\boldsymbol{c}\big)_i = \big(\boldsymbol{1}\big)_i = 1.$$

However, this can be done only under the assumption $c_i > 0$, as if $c_i = 0$ the gradient might not be defined (as $\boldsymbol{c} \geq 0$). So, for $i = 1, \ldots, m$, we either have $c_i = 0$ or $\sum_{j=1} c_j y_i y_j \boldsymbol{x}_i^\top \boldsymbol{x}_j = 1$.
Recalling from Solution 2.2 that $\boldsymbol{w} = \sum_{i=1}^m c_i y_i \boldsymbol{x}_i$, we obtain that

$$\begin{aligned}
\boldsymbol{w}^\top \boldsymbol{w} &= \sum_{i=1}^m \sum_{j=1}^m c_i c_j y_i y_j \boldsymbol{x}_i^\top \boldsymbol{x}_j \\
&= 0 + \sum_{i:c_i>0} c_i \sum_{j=1}^m c_j y_i y_j \boldsymbol{x}_i^\top \boldsymbol{x}_j \\
&= \sum_{i:c_i>0} c_i \cdot 1 \\
&= \sum_{i=1}^m c_i
\end{aligned}$$

which concludes the proof.